

Lab Exercise 7: Find All Permutations of a Smaller String Within a Larger String

Objective

To implement a Java program that identifies all occurrences of permutations (anagrams) of a smaller string within a larger string.

Aim

To find and print all starting indices where any permutation of the smaller string occurs in the larger string.

Theory

This problem tests understanding of **string manipulation**, **frequency counting**, and **sliding window algorithms**.

The key idea is to use two frequency arrays (or maps):

- One for the smaller string (pattern).
- One for each substring of the larger string of the same length.

If both frequency arrays match, that substring is a valid permutation of the pattern.

Algorithm / Procedure

1. Input two strings:
 - text (larger string)
 - pattern (smaller string)
 2. Compute character frequency of the pattern.
 3. Slide a window of the same size as pattern across text.
 4. For each window:
 - Compare its frequency with that of pattern.
 - If they match, print the starting index or substring.
 5. Continue until the end of the text string.
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Program Code: Problem2_FindPermutations.java

```
import java.util.*;
```

```
public class Problem2_FindPermutations {
```

```

// Function to find all starting indices of permutations of s in b
public static List<Integer> findPermutations(String s, String b) {
    List<Integer> indices = new ArrayList<>();
    // TODO: Implement your logic here
    return indices;
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter smaller string: ");
    String s = sc.next();
    System.out.print("Enter bigger string: ");
    String b = sc.next();
    List<Integer> result = findPermutations(s, b);
    System.out.println("Starting indices of permutations:");
    for (int index : result) {
        System.out.print(index + " ");
    }
    System.out.println();
    sc.close();
}
}

```

Sample Input / Output

Input:

cbaebabacd

abc

Output:

Permutation found at index: 0

Permutation found at index: 6

Explanation:

- At index 0 → substring "cba" is a permutation of "abc".
- At index 6 → substring "bac" is also a permutation of "abc".

Evaluation Criteria (Total Marks: 10)

Test	Description	Marks
Program compiles and runs successfully	No errors	2
Correct logic for permutation detection	Matches expected output	3
Efficient sliding window implementation	Avoids redundant checks	2
Input/output format	Matches specification	2
Code clarity and comments	Readable and well-structured	1
Total		10

Result

The program successfully identifies all permutations of a smaller string that appear within a larger string.